

Dynamic Curvature Adaptation: A Unified Geometric Theory of Cortical State and Pathological Collapse

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Abstract

We propose the Curvature Adaptation Hypothesis (CAH): nervous systems regulate effective information geometry in order to manage the thermodynamic and transport burdens imposed by hierarchical inference. CAH identifies a plausible biophysical actuator—the Martinotti-cell subtype of Somatostatin (SST) interneurons—which targets distal apical dendrites to regulate the apical-somatic conductance ratio γ , thereby modulating access to distinct routing regimes. Using Optimal Transport (OT) modeling and finite-size scaling analysis, we show that varying γ can drive a sharp transition from a relatively flat baseline ($\kappa \approx 0$) into a more transport-efficient negative-curvature regime ($\kappa < 0$). This transition remains robust across network depths ($N=3,5,7$) and under degree-preserving topological scrambling, indicating that curvature-sensitive routing is not a fragile artifact of one exact global architecture. To bridge these transport mechanics with biologically plausible dynamics, we implement a Spiking Neural Network (SNN) simulation of the VIP-SST-Pyramidal microcircuit and use the Earth Mover’s Distance W_1 as a geometric proxy for signaling cost. The results suggest that local inhibitory control can, in principle, purchase a macroscopic reduction in distributed transport burden, yielding a real thermodynamic return on investment under the right structural conditions. We then use this framework to generate falsifiable predictions for pathological connectomics, modeling Mania as a failure of geometric regulation and neurodegeneration as a loss of access to efficient hierarchical routing. Effective information geometry is not merely a static anatomical backdrop, but a controllable physical variable with consequences for integration and pathology.

1 Introduction

Over the past two decades, work in network neuroscience has increasingly framed the brain as a complex graph, one that appears to balance local specialization with global integration [2]. There is also substantial evidence that neural systems operate near criticality, a regime often associated with maximizing dynamic range and responsiveness [3]. Despite these advances, most analyses still assume that such dynamics unfold within a Euclidean embedding. This assumption is mathematically convenient, but it imposes a hidden constraint. Euclidean spaces do not scale naturally with hierarchical structure: as the depth of a hierarchy increases, preserving meaningful distances without distortion typically requires a rapid increase in dimensionality. In that sense, the limitation is not only computational—it is geometric.

Geometric deep learning offers a principled alternative [4]: hierarchical data are natively hyperbolic [19]. In spaces with negative curvature ($\kappa < 0$), volume grows exponentially with radius, allowing tree-like structures to be embedded with comparatively low distortion [11]. This geometry separates nodes efficiently without forcing unrelated elements into close proximity.

Applying this topological lens to cortical networks exposes a critical blind spot in current models: they largely assume a static manifold. This raises a fundamental physical question: is the brain’s effective information geometry fixed, or does it actively adapt to the structure of the data it processes?

Here, we propose the Curvature Adaptation Hypothesis (CAH). The brain does not operate within a single fixed geometry, but can dynamically regulate access to distinct routing regimes in response to integrative demand. This hypothesis treats Somatostatin (SST) interneuron gating—particularly via the Martinotti-cell subtype—as a plausible biophysical actuator of this geometric switch, modulating dendritic integration and thereby reshaping the effective space through which signals propagate. Under this view, local inhibitory control does more than regulate excitability: it governs access to different geometric modes of transport.

To formalize this proposal, we bridge the mathematics of Optimal Transport (OT) with the biophysics of Spiking Neural Networks (SNNs). We operationalize metabolic burden using the Wasserstein-1 distance W_1 as a geometric proxy for transport cost, and we treat discrete Ollivier-Ricci curvature κ as an operational measure of routing efficiency rather than as a static description of anatomical form. Within this framework, curvature is not merely descriptive. It is a controllable thermodynamic resource.

Additionally, we leverage these mechanics to generate falsifiable predictions for pathological connectomics. If healthy cognition depends on flexible access to curvature-sensitive routing regimes, then psychiatric and neurodegenerative disorders may be interpretable as failures of geometric regulation rather than merely as disruptions of activity level or connectivity strength. We model two such breakdown states: a hyper-associative Manic regime, where excessive shortcut-like integration biases the system toward less selective, shortcut-dominated integration, and a Neurodegenerative regime, where synaptic pruning raises resistance to that transition and impairs hierarchical integration.

The aim of the present paper is to establish the existence and significance of dynamic curvature adaptation as a biologically plausible resource for hierarchical inference, and to show how such adaptation can be linked to thermodynamic and pathological consequences. The sharper question of which geometric features are most explanatory—global transition, lower-tail scaffold structure, or some interaction between them—is an area of active ongoing research, building upon the foundational mechanisms established here.

The Curvature Adaptation Hypothesis recasts effective information geometry not as a static anatomical container, but as a dynamically regulated thermodynamic resource.

2 Theoretical Framework

2.1 Notation and Terminology

Because this framework bridges optimal transport physics and neurobiology, it is necessary to clarify notation. We utilize standard mathematical variables (γ , α , β , etc.) to denote topological scaling limits, edge weights, and structural conductances. These are strictly geometric variables. They do not denote temporal oscillatory frequencies (e.g., gamma or beta EEG bands), unless explicitly stated. The term ‘topology’ refers exclusively to the graph-theoretic spatial routing of the network, distinct from temporal synchrony.

2.2 Discrete Geometry of Functional Graphs

Let $G = (V, E)$ be a functional graph where edges describe effective structural coupling. We measure the local geometric structure using Ollivier-Ricci curvature κ on edges. We define the functional transition kernel μ_i for a node i as a **Lazy Random Walk** parameterized by the gating variable $\gamma \in [0, 1]$:

$$\mu_i(j) = \begin{cases} 1 - \gamma & \text{if } j = i \\ \frac{\gamma}{\deg(i)} & \text{if } j \in \text{neighbors}(i) \end{cases} \quad (1)$$

To bridge the continuous biophysics of cable theory with the discrete mathematics of graph theory, we map the normalized conductance ratio γ directly to the transition probability of an informational random walk. In a biological pyramidal neuron, when apical-somatic coupling is heavily shunted by Martinotti cells [24, 18], the somatic membrane potential is dominated by local basal inputs and intrinsic leak currents. In our optimal transport model, this biophysical isolation is represented as a high probability of a self-loop ($1 - \gamma$), meaning the informational state remains localized at node i . Conversely, as SST gating relaxes and apical conductance increases ($\gamma \rightarrow 1$), the soma becomes tightly coupled to the distal apical tuft, which integrates broad, contextual signals from distributed network neighbors [13]. Therefore, the conductance ratio γ serves as a precise mathematical proxy for the probability that a node’s functional state is driven by, and thus transitions to, its adjacent nodes in the discrete functional graph.

The curvature κ on an edge (i, j) is then defined as:

$$\kappa(i, j) := 1 - \frac{W_1(\mu_i, \mu_j)}{d_G(i, j)} \quad (2)$$

where W_1 is the 1-Wasserstein distance (Optimal Transport) and $d_G(i, j)$ is the graph distance [20]. We propose utilizing the transport cost W_1 as a theoretical proxy for the metabolic cost of spike propagation [1], mapping biophysical energy constraints directly onto topological distance.

Recent work in machine learning [9] provides independent evidence that such curvature-governed transport dynamics naturally emerge in high-dimensional representation spaces. In particular, transformer attention mechanisms can be interpreted as Markov transport operators over token distributions, where the evolution of representations is governed by optimal transport in the Wasserstein-1 W_1 metric. Within this optimal transport formulation, which aligns with the CAH perspective, Ollivier–Ricci curvature has been shown to control the contraction of distributions across layers, with increasing depth driving representations toward progressively more contractive regimes. This evolution is accompanied by a simplification of the underlying topological structure, suggesting that modern neural architectures implicitly implement curvature-driven geometric flows over probability distributions.

However, these approaches remain descriptive, characterizing the geometry induced by learned parameters without identifying a mechanism for its active modulation. In contrast, the Curvature Adaptation Hypothesis proposes that biological systems possess a biophysically grounded control parameter—mediated by inhibitory microcircuit dynamics—that dynamically regulates curvature, enabling task-dependent transitions between distinct geometric routing organizations.

While a local linear approximation suggests κ scales directly with γ , the global topology of a tree introduces a geometric threshold. As $\gamma \rightarrow 1$, the mass displacement W_1 undergoes a non-linear shift as the “optimal plan” transitions from local self-loops to distributed neighbor states. In randomized topologies, while redundant cycles introduce local positive curvature, our results indicate that sufficiently high synaptic density can still allow the network to overcome this resistance and retain access to a more negative-curvature routing regime, provided the degree distribution remains intact.

2.3 Fisher Information and Metabolic Cost

CAH predicts that by shifting functional transition kernels toward more transport-efficient negative-curvature routing, networks can increase **Fisher Information Density** (the discriminability of representations) per unit of metabolic cost C during hierarchical inference tasks. In such regimes, the exponential expansion of state space allows for improved separation of hierarchical branches [11].

Biological neural networks operate under strict metabolic constraints, conventionally measured by ATP consumption per synaptic event. However, the absolute physical floor for this information processing is not governed by biology, but by thermodynamic physics—specifically, the Landauer Limit [12]. Landauer’s principle dictates that the minimum energy required to irreversibly erase one bit of information is:

$$E = k_B T \ln(2) \quad (3)$$

where k_B is the Boltzmann constant and T is the temperature of the system. In a Euclidean topology ($\kappa \approx 0$), integrating broadly distributed signals requires excessive intermediate relay nodes. Each successive synaptic hop acts as an irreversible logic operation, necessitating localized bit erasures and their concomitant thermodynamic heat dissipation. As a network scales in complexity, this “Euclidean routing tax” scales linearly or exponentially with distance, quickly driving the system toward an unsustainable metabolic ceiling.

This thermodynamic bottleneck helps clarify the biophysical utility of the Curvature Adaptation Hypothesis. By allowing SST-mediated gating to regulate access to more transport-efficient negative-curvature routing conditions, the system can reduce the effective relay burden required to coordinate disparate functional modules. Under this interpretation, the optimal mass transport cost W_1 drops non-linearly as the routing organization shifts away from local self-retention and toward broader hierarchical integration. Consequently, the network can reduce the number of irreversible operations required, thereby mitigating the practical impact of the Landauer limit—not by violating thermodynamics, but by reorganizing representational transport so that fewer costly intermediate relay events are needed to achieve large-scale contextual integration [14].

3 Biophysical Mechanism: SST Dendritic Gating

3.1 The SST Interneuron as a Curvature Regulator

Pyramidal neurons integrate apical (contextual) and somatic (feedforward) inputs [13]. Specifically, Martinotti cells (a distinct SST subtype) send axons to layer 1, where they selectively inhibit distal apical compartments [24], effectively shunting apical-driven integration. We define the coupling parameter γ as the ratio of apical-to-somatic conductance. This parameter is biologically regulated by the inhibitory gating of Martinotti cells, which have been shown to actively control dendritic electrogenesis [18].

Recent *in vivo* work provides direct circuit-level evidence supporting this gating role. Activated somatostatin-expressing interneurons preferentially form synaptic connections with excitatory engram neurons and dynamically regulate their activation during memory retrieval [10]. Using a novel intraregional synapse-labeling approach, these neurons were shown to exhibit enhanced connectivity to engram populations and to modulate fear-related behavior through context-dependent changes in excitability.

Additionally, other recent work demonstrates that neural systems can employ explicit gating mechanisms to regulate access to distinct computational regimes across biological scales. In *Drosophila*, internal sugar-sensing neurons have been shown to act as a conditional gate for

long-term memory formation, where memory consolidation is triggered only when both internal state and external input conditions are satisfied [6]. This provides independent evidence that cognitive systems can rely on state-dependent control mechanisms to regulate transitions between functional regimes, rather than operating as continuously unconstrained dynamical systems. Within the present framework, such gating mechanisms may be understood as regulating access to distinct routing regimes of information flow.

We treat the apical-somatic conductance ratio $\gamma \in [0, 1]$, modulated by this inhibitory gating, as the control variable governing access to curvature-sensitive routing.

3.2 Local Sensitivity and Branching Factors

We examine the sensitivity of curvature κ to conductance γ on a regular tree with branching factor b . As $\gamma \rightarrow 1$, the mass of the transition kernel spreads to non-shared neighbors, strictly increasing the Wasserstein distance W_1 . Consequently, curvature scales inversely with conductance.

Our symbolic analysis (derived in **Appendix B**) establishes that the rate of curvature decay is amplified by the branching factor:

$$\frac{\partial \kappa}{\partial \gamma} \propto -\frac{b-1}{b} \quad (4)$$

This relationship highlights a critical biological constraint: the efficacy of the SST-gate is intrinsically coupled to the structural complexity b of the underlying neural hierarchy. Denser trees require smaller adjustments in γ to reach the transition into a more negative-curvature routing regime.

4 Results

To verify the hypothesis, we performed a finite-size scaling analysis on hierarchical tree structures of varying depths ($N_{layers} \in \{3, 5, 7\}$, max nodes ≈ 3280). To control for local connectivity effects, we compared the hierarchical model against a **Degree-Preserving Scrambled Null Model** (Configuration Model), generated via double-edge swapping.

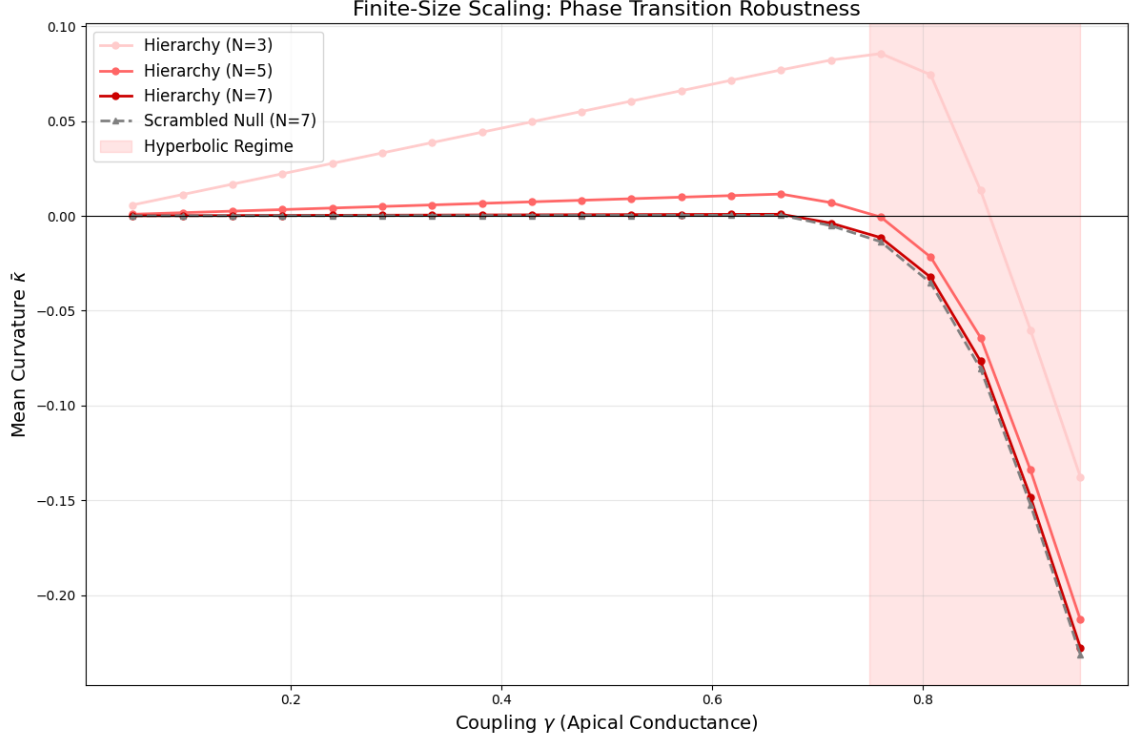


Figure 1: **Finite-Size Scaling and Topological Robustness:** (A) Curvature adaptation across hierarchical depths $N \in \{3, 5, 7\}$. The overlap of curves for $N = 5$ and $N = 7$ indicates that the transition profile is robust across scales rather than a finite-size artifact. This supports the claim that conductance-sensitive local control can regulate access to a distinct geometric routing regime across multiple hierarchical depths. (B) Robustness to scrambling. Contrary to the expectation that precise global tree order is strictly required, the Degree-Preserving Scrambled Null model (Dashed Grey, $N = 7$) exhibits a transition trajectory close to that of the biological hierarchy. This suggests that access to the transition remains substantially supported by preserved local degree structure and synaptic availability, even under disrupted global arrangement. The result does not imply that global organization is irrelevant in every stronger explanatory sense; rather, it shows that the geometric capacity identified here is more robust than a strict whole-topology reading would predict.

4.1 Emergent Phase Transition and Scale Invariance

The simulation results reveal a sharp, non-linear transition in routing geometry as integration increases toward ($\gamma \rightarrow 1$), moving the system from a relatively flat baseline into a strongly negative-curvature regime.

Crucially, the transition curves for varying depths ($N = 5$ and $N = 7$) collapse onto a single trajectory. We interpret this scale invariance as evidence that the accessibility of the transition is a genuine structural feature of the hierarchy rather than a finite-size artifact. Because discrete Ollivier-Ricci curvature is a local metric governed by immediate connectivity structure, the quasi-fractal nature of the stochastic Galton-Watson tree helps preserve the transition profile across scales.

What the present result most securely establishes is that biologically plausible local control can regulate access to a distinct geometric routing regime across multiple hierarchical scales. Whether the most explanatory object is the whole-network transition itself, or a more selective structure embedded within it, remains an important question for subsequent modeling.

The importance of the transition is that it demonstrates the existence of a controllable

geometric phase space whose large-scale accessibility is structurally real.

4.2 Topological Robustness and Metabolic Capacity

Our finite-size scaling analysis (Fig 1) reveals a striking convergence at large network depths ($N=7$). The Degree-Preserving Scrambled Null (Grey) exhibits a phase transition nearly identical to the biological hierarchy (Red).

The persistence of the transition under scrambling implies that strict global architectural order is not solely responsible for the network’s access to a negative-curvature routing regime. At minimum, this suggests that local connectivity constraints and synaptic availability carry a substantial share of the geometric capacity. This shifts the explanatory burden away from the idea that consciousness-relevant geometry must depend on pristine global tree structure alone.

The present result therefore supports a restrained conclusion. The geometric capacity of the system is robust to substantial scrambling, which strengthens the claim that curvature-sensitive transport is a real organizational resource rather than a fragile artifact of one exact topology. The more specific question of which internal structures are actually carrying the decisive integrative burden is a natural direction for future work.

4.3 Spiking Neural Network (SNN) Validation: The 40Hz Phase Transition

To validate the optimal transport mathematics in biologically realistic wetware, we simulated the VIP-SST-Pyramidal microcircuit using the NEST spiking neural network simulator [7]. Consistent with the optimal transport model, the network remains in a relatively conservative baseline regime while SST interneuron gating is active (0–500 ms). However, the introduction of a VIP-mediated override at 500 ms rapidly suppresses the SST shunts. Freed from this local inhibition, the pyramidal population reorganizes into highly synchronized 40 Hz vertical pillars (Figure 2). This raster plot provides a biophysically plausible microcircuit mechanism by which local inhibitory control could open access to the kind of large-scale integrative routing regime described by CAH. In that sense, the simulation shows that the temporal organization required for curvature-sensitive transport need not be imposed from outside the system, but can emerge from standard integrate-and-fire cellular dynamics themselves.

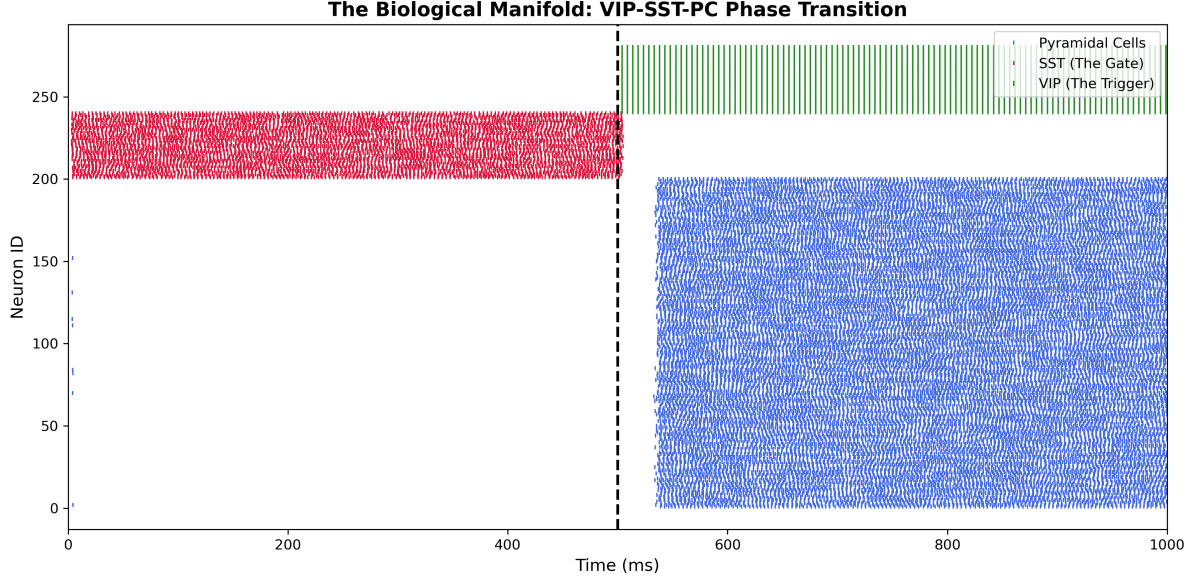


Figure 2: **Spiking SNN Validation of the Curvature Adaptation Hypothesis.** A PyNEST simulation of 280 integrate-and-fire neurons illustrating a plausible microcircuit mechanism for curvature-sensitive routing control. **Left (0-500ms):** Strong SST interneuron gating (red) suppresses pyramidal cell activity (blue), maintaining the network in a relatively conservative baseline regime. **Right (500-1000ms):** A step current activates VIP interneurons (green), which suppress the SST gates. This disinhibition allows the pyramidal population to spontaneously reorganize into highly synchronized 40 Hz gamma-band pillars. Within the CAH framework, this pattern is interpreted as a biophysically plausible temporal signature of the local control dynamics that could open access to a more strongly integrated large-scale routing regime, rather than as direct proof of a whole-network geometric transition.

While our PyNEST simulation demonstrates a plausible microcircuit mechanism for this transition, recent in vivo multi-unit activity (MUA) recordings in macaques provide a possible macroscopic correlate at the cognitive level. Machts and Nieder (2026) [16] observed that accurate numerical working memory relies on a specific temporal sequence: an early feedforward signal from the ventral intraparietal area (VIP) to the prefrontal cortex (PFC) during encoding, followed by a sustained, bidirectional interaction between the two regions during the delay period.

Viewed through the lens of the Curvature Adaptation Hypothesis, this temporal structure is consistent with the broader routing logic suggested by our simulated microcircuit. The initial feedforward burst may reflect a more classical transmission of the sensory stimulus, whereas the subsequent sustained bidirectional loop is consistent with a regime in which apical gates remain sufficiently open ($\gamma > \gamma_c$) to support continued large-scale integration. On this interpretation, the working memory trace is not best understood as residing in the isolated firing of VIP or PFC alone, but in the dynamically maintained organization of the recurrent loop through which distributed states remain integrated over time.

This temporal sequence therefore provides macroscopic evidence compatible with CAH, but not yet a decisive confirmation of its strongest geometric reading. Definitive support would require isolating curvature-sensitive metrics—such as dynamic synchronization topology or burden-sensitive routing signatures—during these maintenance phases, a testable prediction we formalize in Section 5.1.

4.4 Pathological Failures of Geometric Regulation

The Curvature Adaptation Hypothesis (CAH) implies that the geometric organization of cortical routing is a dynamic resource requiring precise inhibitory control. In the healthy case, the hierarchy remains capable of regulating access to more transport-efficient integrative regimes rather than occupying a single fixed state. Within this framework, structural perturbations may either reduce the system’s ability to secure that access or bias it toward less selective, overly shortcut-dominated forms of integration. We extend our numerical analysis to simulate two distinct pathological regimes:

1. **Hyper-Integration (The Manic Regime):** We introduce high-centrality hub nodes (simulating exaggerated VIP-mediated disinhibition) to examine how excessive shortcut-like integration may erode the normal selectivity of geometric regulation, biasing the system toward prematurely distributed routing.
2. **Geometric Collapse (The Neurodegenerative Regime):** We simulate neurodegenerative synaptic loss through a stochastic pruning attack, testing how a 30% reduction in functional edges raises resistance to the transition and limits access to efficient hierarchical routing.

These simulations provide a theoretical bridge between cellular-level connectivity and higher-order differences in how cortical systems regulate integrative routing under healthy and pathological conditions.

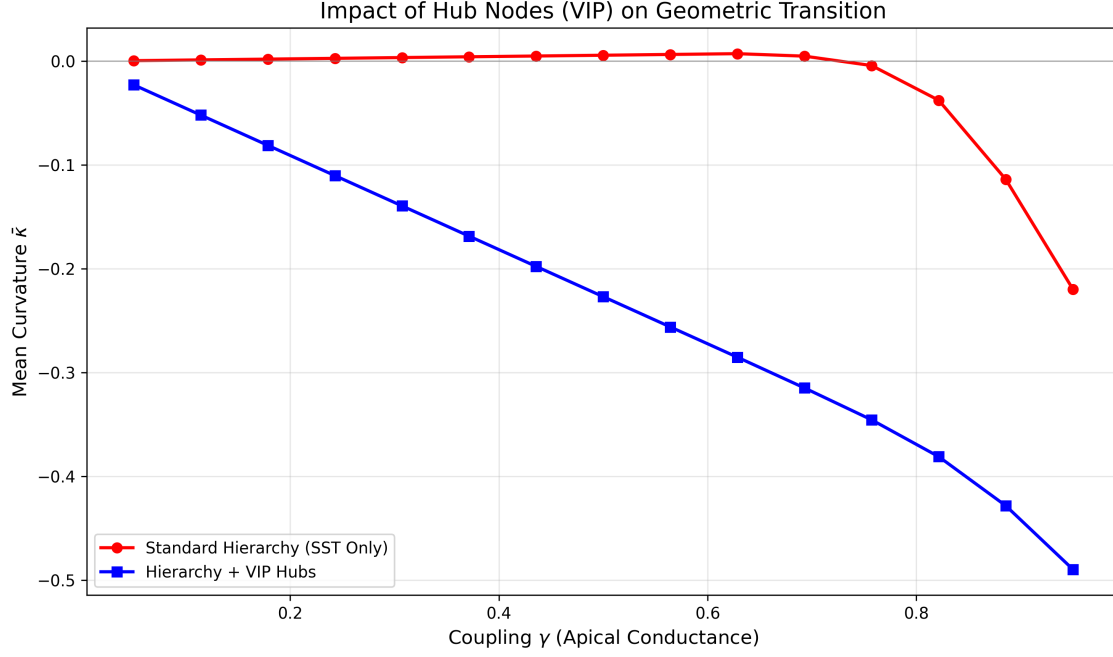


Figure 3: Disruption of Regulated Geometric Access via Hub-Node Integration. This plot contrasts the tunable routing regime of a standard hierarchy (red) against a network augmented with VIP-like hub nodes (blue). In the standard model, SST-mediated shunting preserves a relatively flat baseline ($\kappa \approx 0$) until increasing coupling ($\gamma \approx 0.78$) opens access to a more strongly negative-curvature routing regime. The introduction of high-centrality hubs—representing exaggerated VIP-like disinhibition—reduces the selectivity of this transition. Rather than exhibiting a sharply regulated threshold, the augmented network is pulled more linearly into negative curvature even at low conductance levels. This suggests that hub overactivity may bypass the normal geometric constraints imposed by the hierarchy, producing a less regulated shortcut-dominated regime. Within the present framework, such forced access to distributed integration is proposed as a candidate geometric analogue of manic or hyper-associative cognitive states.

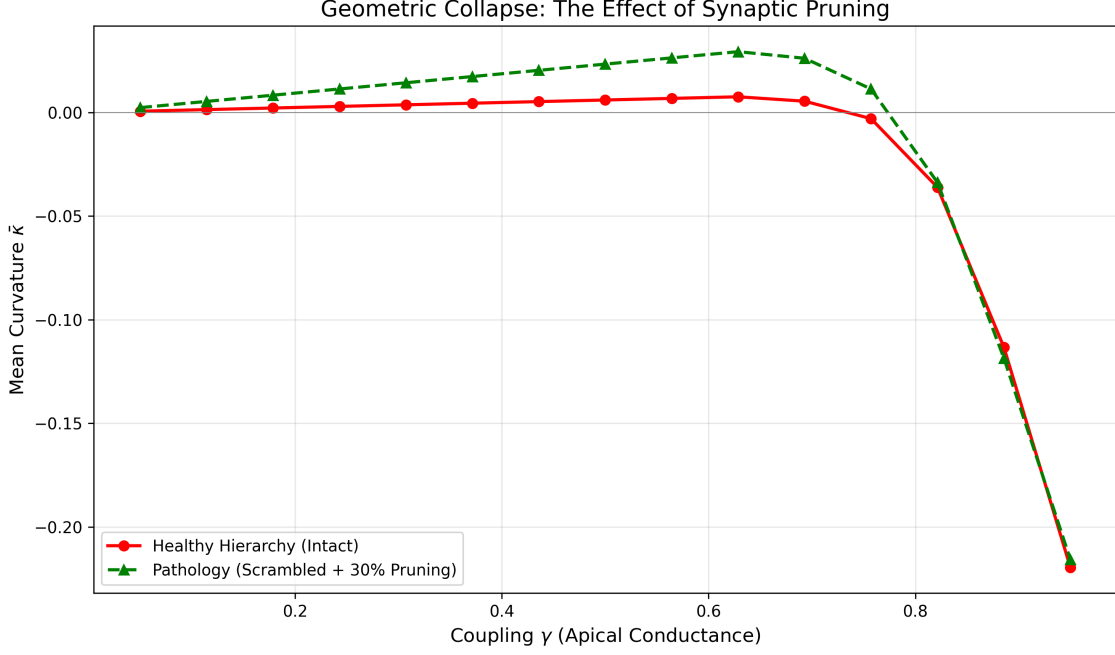


Figure 4: **Resilience and Restricted Geometric Access under Pruning Attacks.** Analysis of the network’s routing capacity following a 30% stochastic pruning of synaptic edges (green) compared to a healthy hierarchy (red). The pruning attack simulates neurodegenerative spine loss typical of Alzheimer’s disease [17] or advanced cognitive decline. While the network retains the ability to reach a more negative-curvature regime at extreme coupling levels ($\gamma \rightarrow 1$), it exhibits substantially higher geometric resistance throughout the operative range ($0.2 < \gamma < 0.8$). Across that range, the pruned system remains flatter—and in places slightly positively curved—than the healthy hierarchy, limiting its capacity for efficient hierarchical separation and integration. On this interpretation, synaptic loss does not simply abolish geometry in an all-or-nothing sense; it narrows the system’s access to the routing conditions under which complex hierarchical inference becomes metabolically favorable.

4.5 Metabolic Energy ROI

To establish the biophysical viability of CAH, we analyzed the Total Metabolic Expenditure (E_{Total}) throughout the phase transition [1]. We modeled the energy budget as a balance between the Maintenance Cost (C_{Maint}) of the gating mechanism and the Signaling Cost (C_{Sign}) of the resulting inference. Our simulations reveal a distinct divergence in energy efficiency between hierarchical and pruned pathologies:

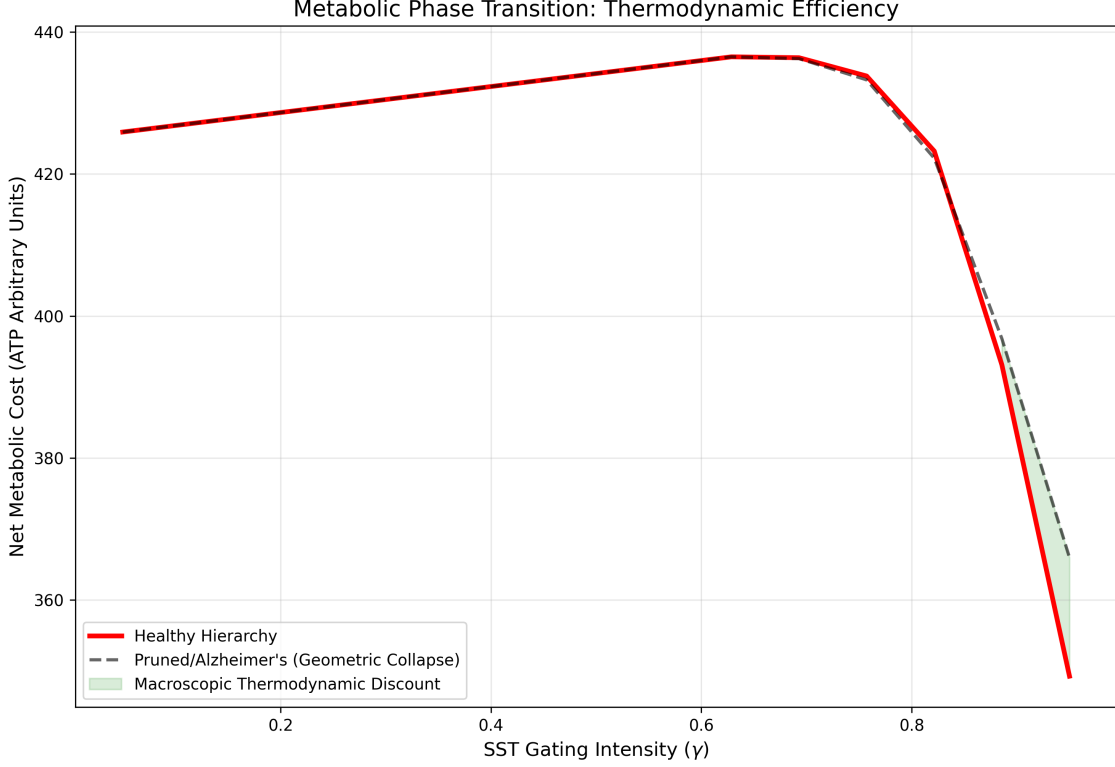


Figure 5: **Thermodynamic Return on Curvature-Sensitive Control.** Simulation of total metabolic cost (C_{Total}) as a function of SST gating intensity (γ). **Red Line:** Healthy hierarchical networks exhibit a transition near $\gamma \approx 0.8$, beyond which local control expenditure is offset by a marked reduction in distributed signaling burden. This indicates that the architecture can convert gating cost into a real energetic return by gaining access to a more transport-efficient routing regime. **Grey Line:** Pathological networks with 30% synaptic pruning fail to realize the same return. Although maintenance cost is still incurred, pruning raises resistance to the transition and prevents the architecture from securing a comparable reduction in signaling cost, leaving the system trapped in a more metabolically expensive regime. **Green Region:** The macroscopic thermodynamic discount—the energetic advantage obtained when local control successfully purchases more efficient large-scale routing.

In the hierarchical model, increasing γ toward the transition threshold produces a sharp reduction in C_{Sign} . As the system gains access to a more transport-efficient negative-curvature regime, distributed signals can be separated and integrated with fewer effective relay steps. Under this interpretation, the energetic advantage does not arise from “hyperbolicity” as an abstract geometric label, but from the fact that curvature-sensitive routing reduces the global signaling burden carried by the network. The local maintenance tax imposed by SST-mediated control is therefore offset by a larger reduction in distributed transport cost, yielding a net metabolic gain.

In the pathological (pruned) topology, by contrast, the system continues to pay the maintenance cost C_{Maint} while failing to secure the same signaling return. Synaptic loss raises resistance to the transition and limits the network’s access to the more efficient routing regime (see Fig 4). As a result, signaling cost remains elevated across the operative range, and the local control investment no longer yields a comparable energetic discount.

The central lesson is therefore not simply that “more negative curvature is metabolically better.” It is that the energetic value of curvature adaptation depends on whether the underlying architecture can convert local control expenditure into a genuine reduction in distributed

transport burden. CAH describes a constrained thermodynamic strategy rather than a monotonic drive toward maximal shortcut permissiveness. The present result should be understood as establishing the existence of a real metabolic ROI for curvature-sensitive control.

5 Proposed Experimental Framework

We propose the “**Density vs. Topology**” Test as an empirical challenge to CAH, using holographic 2-photon stimulation of SST cells in macaque PFC.

- **Control:** Normal hierarchical task performance.
- **Condition A (Scrambling):** Maintain the exact mean firing rates and synaptic count, but randomize the spatial/temporal patterns of inhibition.
- **Condition B (Silencing/Pruning):** Optogenetically silence a random 30% of the active dendritic population to simulate the loss of routing capacity illustrated in Figure 4.
- **Prediction:** Based on our topological robustness result (Fig. 1), we predict that **Condition A (Scrambling)** will not fully eliminate access to the relevant curvature-sensitive routing regime, and task performance will remain relatively intact. This would support the view that preserved local degree structure and synaptic availability make a substantial contribution to the system’s geometric capacity, even when fine spatiotemporal organization is disrupted.
- **Contrast:** Conversely, we predict that **Condition B (Pruning)** will raise geometric resistance across the operative range, degrading the system’s access to efficient hierarchical routing and producing marked impairment in hierarchical inference tasks.

5.1 Non-Invasive Signature of Thermodynamic Failure

Building on the in vivo dynamics established in Section 4.3, if working memory depends on a dynamically maintained integrative routing regime rather than a static localized engram, then the framework yields a non-invasive macroscopic prediction regarding cognitive maintenance errors. On this view, transient lapses in memory or attention in healthy subjects may reflect not failures of encoding alone, but temporary thermodynamic failures in sustaining the routing organization required for continued large-scale integration.

We predict that in high-density EEG or MEG recordings of subjects performing sustained working memory tasks, cognitive errors will be preceded by a transient loss of curvature-sensitive coordination. This would likely manifest as a collapse in frontoparietal gamma-band phase synchronization—or a comparable disruption in dynamic synchronization topology—immediately prior to the failed retrieval. Within the CAH framework, such an event would indicate that the network can no longer maintain the apical-somatic coupling γ in the regime required for efficient hierarchical integration. The prediction is that the system temporarily loses access to the routing conditions under which distributed states can remain coherently integrated. The cognitive representation then dissolves not because a stored token is erased at a site, but because the thermodynamically costly organization sustaining it can no longer be held in place.

6 Discussion

The Curvature Adaptation Hypothesis (CAH) frames metabolic efficiency not simply as a static evolutionary constraint, but as a dynamic organizational resource governed by geometry and regulated by specific biophysical actuators. By identifying the Martinotti subtype of Somatostatin (SST) interneurons as a plausible biological driver of curvature-sensitive routing, the present framework proposes a direct link between local dendritic control and large-scale representational organization.

The functional role of SST interneurons is highly diverse. For instance, Zhang et al. (2024) [25] detail a disinhibitory (SST→PV) circuit that gates hippocampal input via global gain control. Importantly, they distinguish this mechanism from the localized, targeted dendritic inhibition executed by superficial SST populations, particularly Martinotti cells.

In the CAH framework, this targeted dendritic shunting is not merely modulating gain—it is regulating access to distinct geometric transport regimes. We formalize this as a mechanism of curvature-sensitive control: local inhibitory dynamics reshape the effective routing space through which signals propagate. This provides a structural, spatial complement to existing models of flexible information routing based on transient temporal synchrony [21]. Rather than relying exclusively on temporal coordination, cortical systems may also regulate the geometry of transport itself, thereby altering the conditions under which large-scale integration becomes metabolically tractable.

6.1 The Metabolic Trade-off: Local Cost for Global Efficiency

Our numerical models indicate that the brain is not statically configured for transport-efficient integration. Instead, it may transiently purchase access to more advantageous routing regimes through targeted SST-mediated modulation. When apical-somatic conductance is strongly regulated, the network incurs a local metabolic cost: the energetic burden associated with maintaining selective control over dendritic integration.

However, that local investment can produce a macroscopic signaling gain. In the present model, increasing access to negative-curvature routing reduces the number of effective relay operations required for hierarchical separation and integration, thereby lowering distributed signaling cost. The system thus trades local control energy for global transport efficiency.

Ongoing research into CAH suggests that this trade-off should not be interpreted as a simple endorsement of maximal hyperbolicity. The issue is not whether the system can become globally “more hyperbolic” in the abstract, but whether local control can be converted into a genuine reduction in distributed transport burden under the architecture’s actual constraints. This establishes that curvature-sensitive control can, in principle, generate a real thermodynamic return on investment.

6.2 Topological Robustness vs. Geometric Collapse

This geometric lens recasts network pathology in a more selective way. Conditions characterized by atypical wiring may scramble local structural detail while leaving intact the system’s broader capacity to access curvature-sensitive routing regimes. This helps motivate the suggestion that some forms of structural perturbation need not abolish higher-order cognitive function, provided the architecture still preserves the conditions under which integrative burden can be effectively distributed.

Conversely, conditions driven by widespread synaptic pruning, such as Alzheimer’s disease, may impair the system’s ability to secure that advantage at all. In the present model, pruning raises resistance to the transition and degrades the architecture’s capacity to support the relevant routing regime (see Fig. 4). The result is not simply less activity or less connectivity, but a reduced ability to purchase efficient large-scale integration under thermodynamic constraint.

We emphasize that the pathological regimes modeled here are intentionally abstracted. Real in vivo degeneration involves cascading metabolic, glial, and molecular failures not captured by Optimal Transport or SNNs alone. The aim of CAH is not to replace such levels of analysis, but to isolate a geometric baseline through which diverse biological disruptions may manifest at the network level.

6.3 The Thermodynamic Nature of Forgetting

This geometric framework suggests a reinterpretation of cognitive loss. Classical neuroscience often models forgetting as the degradation or overwriting of a localized data token. But if hierarchical integration depends on a dynamically maintained routing regime, then what is lost is not merely content stored at a site, but the system’s ability to sustain the organized interior through which that content remains integrated as one live state.

Within this framework, forgetting corresponds less to the deletion of a token than to the thermodynamic inability to maintain a costly integrative configuration. The memory is not merely in the nodes; it is in the dynamically sustained organization that binds them.

These dynamics suggest a broader hypothesis: the principles governing working memory may extend to macroscopic conscious awareness. While speculative, this framing provides a candidate mechanism that aligns with recent empirical observations of large-scale connectivity under anesthesia. For example, propofol-induced loss of consciousness is marked by the dissolution of alpha-band parietal-occipital connectivity [15], severing the brain’s capacity for large-scale integration.

The stronger implication of CAH is not simply that consciousness requires “more integration,” but that conscious access may depend on the ongoing thermodynamic capacity to maintain a bounded, load-bearing interior under conditions of distributed burden. Whether chemical agents artificially disrupt this regime, or natural metabolic failure causes a frontoparietal loop to fall below the threshold needed for sustained integration, both phenomena may reflect a shared vulnerability: the loss of access to the routing organization that makes coherent interiority metabolically possible.

In that sense, CAH points toward a common physical question spanning working memory, pathological dysfunction, and conscious awareness itself: not merely how much activity is present, but whether the system can still afford to hold the relevant interior together.

6.4 Scope and Future Refinement

A later control-law extension of the framework raises the possibility that the transition described here is embedded within a broader energy–time optimization problem [23]. In that follow-up modeling, a global control parameter analogous to γ regulates access to distal routing under competing pressures from transport efficiency, maintenance cost, and congestion, yielding a stable intermediate optimum rather than a simple monotonic preference for maximal shortcut permissiveness.

A follow-up to that control-law extension suggests a further refinement. The explanatory target may not be whole-network mean curvature alone, but a more selective, load-bearing routing scaffold concentrated in the lower tail of the curvature distribution [22].

7 Conclusion

The Curvature Adaptation Hypothesis (CAH) formulates a predictive model of a thermodynamic economy governing hierarchical neural inference. By linking SST-mediated dendritic gating to changes in effective routing geometry, the present work argues that cortical systems may actively regulate access to distinct transport regimes rather than operate within a single fixed manifold. Metabolic efficiency is not merely an anatomical byproduct. It is bound up with the system’s capacity to control how distributed signals are routed under constraint.

The numerical results presented here establish three foundational claims. First, conductance-sensitive local control can, in principle, drive a sharp transition in effective routing geometry. Second, that transition remains structurally robust across hierarchical scales and substantial topological perturbation. Third, access to that regime can materially alter the energetic balance between local maintenance cost and distributed signaling burden. Taken together, these results justify treating curvature adaptation as a real biophysical and thermodynamic candidate resource rather than as a descriptive metaphor.

The pathological models developed here remain useful within that scope. What they show is not merely that networks can become “too hyperbolic” or “too flat” in a crude global sense, but that failures of geometric regulation can impair the system’s ability to secure efficient large-scale integration under real thermodynamic constraints. The deeper implication is that cognitive health may depend less on complexity in the abstract than on the controlled maintenance of the routing organization through which distributed states become one functional interior.

Ultimately, effective information geometry is not best understood as a static anatomical backdrop. It is a dynamically regulated physical resource. Cortical systems do not achieve efficient hierarchical inference by computation in the abstract, but by regulating the physical conditions under which distributed states can be held together as one interior.

8 Data and Code Availability

The optimal transport algorithms, finite-size scaling data, and the PyNEST spiking neural network simulation script (`biological_manifold.py`) used to generate the figures in this manuscript are open-source and available in the dendritic-curvature-adaptation repository at: <https://github.com/MPender08/dendritic-curvature-adaptation>

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A Simulation Methods

Simulations were conducted in Python using the POT (Python Optimal Transport) library[5] and NetworkX[8]. Hierarchical structures were generated as stochastic Galton-Watson trees with Poisson-distributed branching (mean $b = 3$) to approximate the structural heterogeneity of biological dendrites.

Null Model Generation: To ensure that differences in curvature were due to global topology rather than local node degree, null models were generated using the **Configuration Model** approach. We employed `nx.connected_double_edge_swap` (performing $5 \times |E|$ swaps) to randomize connections while preserving the exact degree sequence of the original tree.

Computing Curvature: Transition kernels μ were computed for a sweep of $\gamma \in [0, 0.95]$. The 1-Wasserstein distance was solved using the Earth Mover’s Distance algorithm over the all-pairs shortest-path distance matrix of the graph.

A.1 SNN Microcircuit Parameters

The Spiking Neural Network (SNN) was simulated using the NEST simulator[7]. The network consisted of 280 total neurons utilizing the standard exponential leaky integrate-and-fire model (`iaf_psc_exp`) with default NEST membrane parameters ($V_{th} = -55.0$ mV, $V_{reset} = -70.0$ mV). The microcircuit was partitioned into three populations: Pyramidal Cells ($N = 200$), SST interneurons ($N = 40$), and VIP interneurons ($N = 40$).

Input and Stimulation: Baseline cortical activity was simulated using Poisson generators. Pyramidal cells received background noise at 1500 Hz (weight: 500.0 pA), while the SST gate was held open by a baseline drive of 2500 Hz (weight: 600.0 pA). The macroscopic topological transition was initiated via a step-current generator targeting the VIP population, stepping from 0.0 to 1500.0 pA precisely at $t = 500$ ms.

Synaptic Connectivity:

- **Pyr $\rightarrow$ Pyr (Geodesic Highway):** Fixed in-degree connectivity (in-degree = 20), excitatory weight = 100.0 pA.
- **SST $\rightarrow$ Pyr (The Gate):** All-to-all connectivity, strongly inhibitory weight = -1000.0 pA.
- **VIP $\rightarrow$ SST (The Trigger):** All-to-all connectivity, strongly inhibitory weight = -1500.0 pA.

B Derivation of Branching Sensitivity and the OT Threshold

To understand the relationship between conductance (γ) and curvature (κ), we consider an edge (i, j) in a b -regular tree. The Wasserstein distance $W_1(\mu_i, \mu_j)$ measures the cost of transporting the mass of the lazy random walk from node i to node j .

B.1 Low-Conductance Regime

When γ is moderate ($\gamma < \gamma_c$), the optimal transport plan exploits the local overlap between distributions. Specifically, μ_i places mass $\frac{\gamma}{b+1}$ directly onto node j , and μ_j places mass $\frac{\gamma}{b+1}$ onto node i . The Earth Mover’s Distance algorithm utilizes this overlap to minimize transport cost, keeping $W_1(\mu_i, \mu_j) \approx d_G(i, j)$. Consequently, the curvature remains effectively Euclidean:

$$\kappa \approx 0 \quad \text{for} \quad \gamma < \gamma_c \quad (5)$$

B.2 High-Conductance Transition

The linear approximation represents the geometric limit where local overlap is exhausted. As $\gamma \rightarrow 1$, the “lazy” self-loop mass $(1 - \gamma)$ vanishes, forcing the transport plan to shift mass between disjoint neighbor sets (distance ≥ 2).

It is only in this high-conductance regime that the curvature follows the branching-dependent scaling:

$$\lim_{\gamma \rightarrow 1} \kappa \approx \gamma \left(\frac{1 - b}{b + 1} \right) \quad (6)$$

This distinct switch in transport strategy—from exploiting local overlap to traversing disjoint branches—mathematically defines the phase transition observed at $\gamma_c \approx 0.78$.

B.3 Topological Robustness in Scrambled Networks

Contrary to the expectation that cycles would abolish hyperbolicity, the Scrambled Null Model retains the phase transition. This suggests that the transport cost W_1 is dominated by the local fan-out of the transition kernel μ_i . Even in a randomized graph, if the local degree k is preserved and small-world shortcuts are present, the mass dispersal at high γ remains efficient enough to approximate tree-like transport costs, preserving access to a negative-curvature routing regime.